



ECE 315 Computer Interfacing

LAB 4: HARDWARE SETUP

Winter 2026

HARDWARE SETUP

The hardware configuration shown in Figure 1 needs to be assembled and verified by you in the lab session. This is possible to do in well under one hour. There are five connected subsystems:

1. The Zybo Z7-10 board, which needs to be prepared so that the motor control signals and digital ground can be connected to the upper sockets of the JC Pmod header.
2. An opto-isolation circuit for the motor control signals, which needs to be assembled on the breadboard.
3. A ULN2003-based driver board, which needs to be connected between the opto-isolation circuit and the 28BYJ-48 stepper motor.
4. An external 6 V DC power supply, which supplies the current to the stepper motor windings.

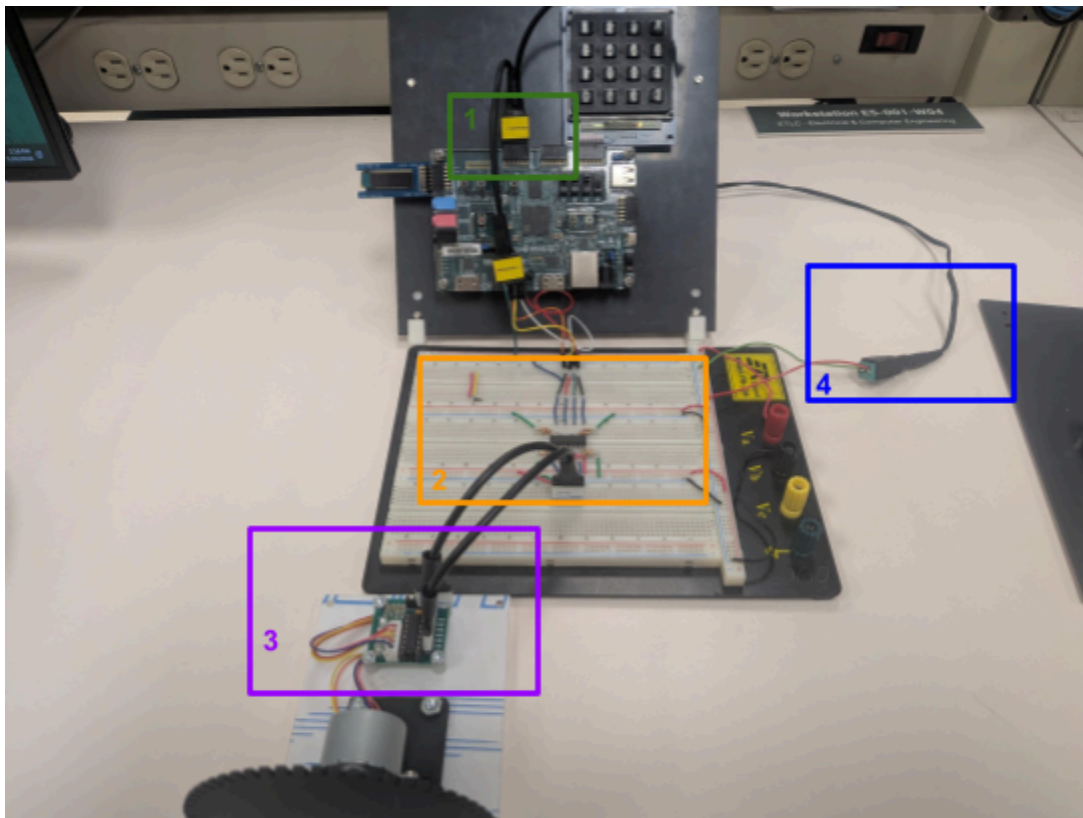


Figure 1: Lab 4 Hardware Configuration



Winter 2026

ECE 315 Computer Interfacing

LAB 4: HARDWARE SETUP

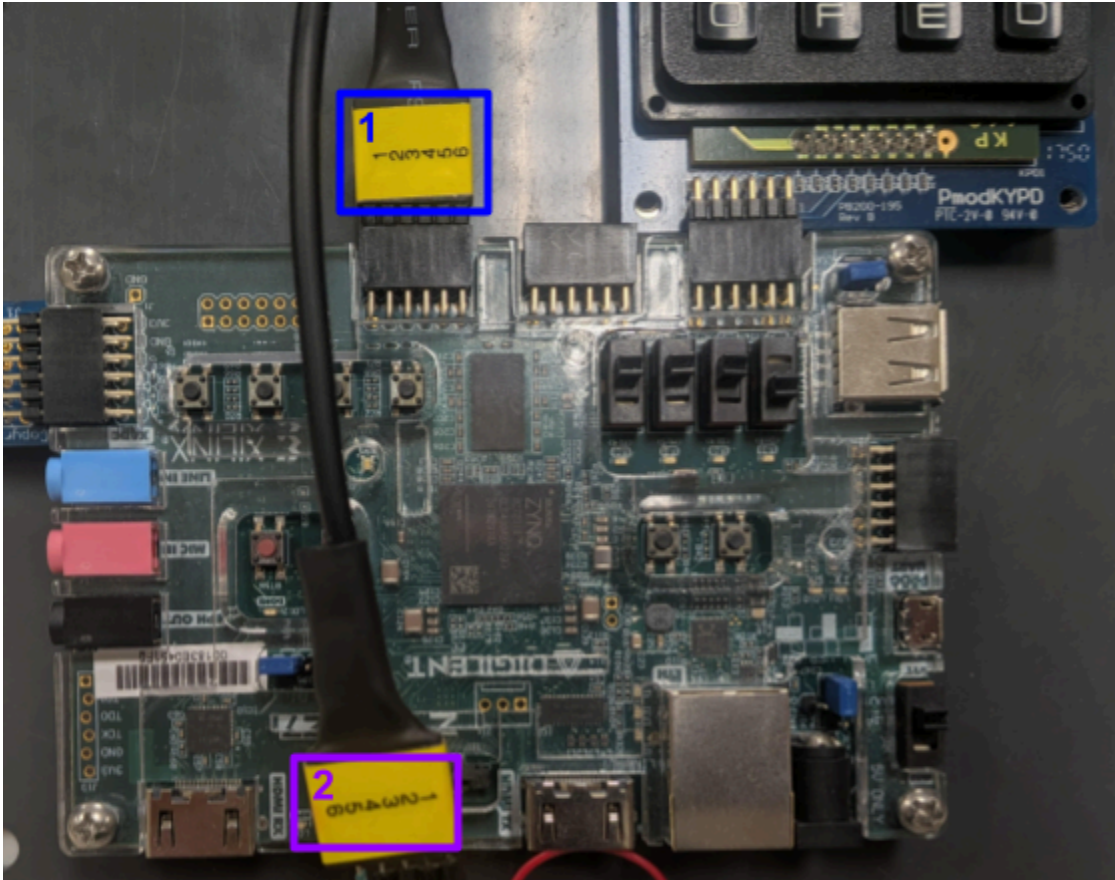


Figure 2: Lab 4 Pmod Port Connection

observe that the Pmod cable includes numbered pin labels. Use these labels as a reference to correctly identify and connect the signals during setup. Refer to Figure 2 for the pin numbering and orientation.

Assemble the opto-isolation circuit on the breadboard:

When the current to the stepper motor's stator windings is switched, voltage spikes are induced across the windings, following Faraday's Law of Induction. These spikes can be many volts in magnitude and can damage the sensitive digital components on the Zybo Z7 board, which operate on a 3.3-V DC supply. Therefore, it is essential that the electrical motor drive control signals from the Zybo Z7 be electrically isolated from the drive transistors and the stepper motor windings. This isolation will be achieved using an opto-isolation circuit, which you will build on the breadboard (see Fig. 3). The grounds on both sides of the optoisolator are different.



Winter 2026

ECE 315 Computer Interfacing

LAB 4: HARDWARE SETUP

Plug the **LTV-847** opto-isolator IC into the breadboard. The small depression in the IC, indicating pin 1, should be at the top-left corner. Connect pins 2, 4, 6, and 8 to ground, each with a separate 220 Ω current-limiting resistor

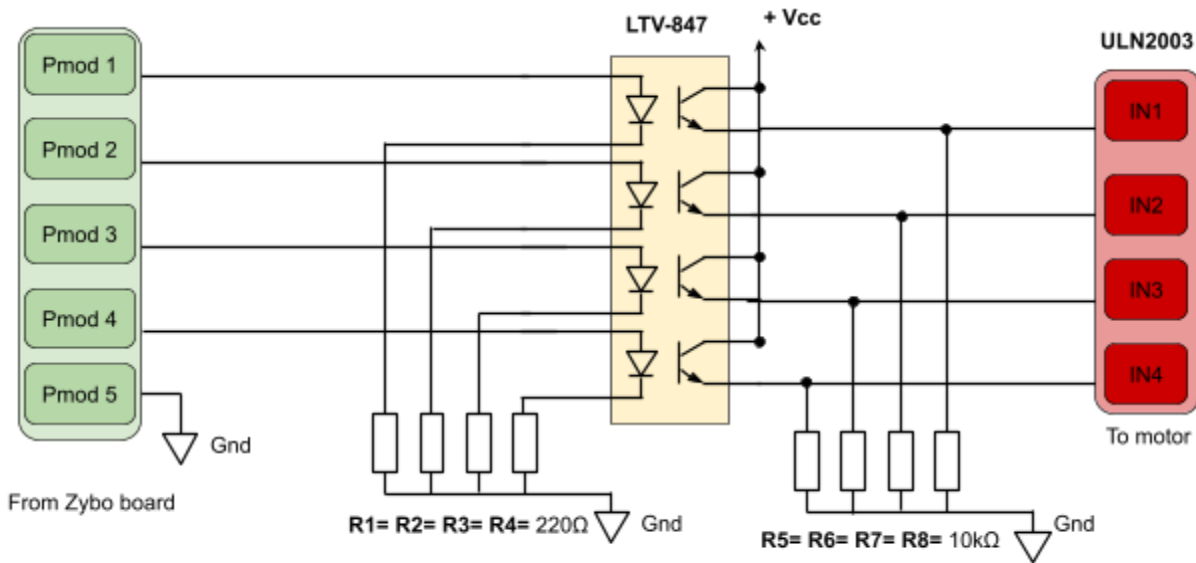


Figure 3: Opto Isolator Circuit Schematic

in series. Connect pins 9, 11, 13, and 15 to a power supply (VCC), each with a separate 10 k Ω pull-down resistor to ground.

Connect the motor control signals and the digital ground rail between the JC Pmod header on the Zybo Z7 board to the breadboard as follows. The JC Pmod header has 12 sockets in two rows as shown in Figure 3, you will be connecting wires to the five rightmost sockets on the top row. The first socket on the top row (going from right to left when looking into the sockets) will have:

1. IN1_in to pin 1 on the LTV-847.
2. IN2_in to pin 3 on the LTV-847.
3. IN3_in to pin 5 on the LTV-847.
4. IN4_in to pin 7 on the LTV-847.
5. Breadboard ground.
6. No connection.



Winter 2026

ECE 315 Computer Interfacing

LAB 4: HARDWARE SETUP

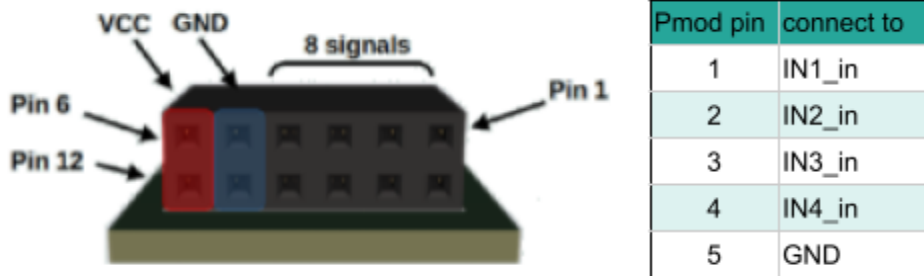


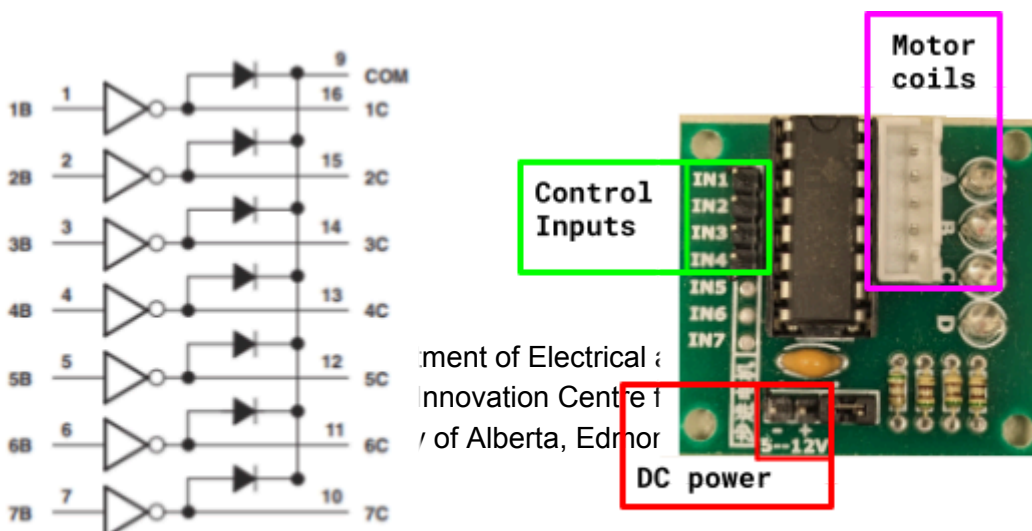
Figure 3: Pmod port pinout (front view)

Connect pins 10, 12, 14 and 16 to the breadboard power rail on the breadboard.

Connect the driver module: The driver board (Fig. 6) is based on the **ULN2003** IC (see Fig. 5), which contains an array of seven pull-down Darlington transistor switches. We will only be using the first four to selectively connect the four outer stepper motor winding terminals to ground. Make the following connections:

1. **IN1_out** from pin 15 on the **LTV-847** to socket **IN1** on the driver module
2. **IN2_out** from pin 13 on the **LTV-847** to socket **IN2** on the driver module
3. **IN3_out** from pin 11 on the **LTV-847** to socket **IN3** on the driver module
4. **IN4_out** from pin 9 on the **LTV-847** to socket **IN4** on the driver module.

Connect the stepper motor: Plug the 5-conductor stepper motor signal cable (Fig. 4) into the driver module socket.



LTV-847	connect to
15	IN1
13	IN2
11	IN3
9	IN4

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Winter 2026

ECE 315 Computer Interfacing

LAB 4: HARDWARE SETUP

You will be using the inexpensive and popular 28BYJ-48 stepper motor (Fig. 5), which is designed to operate using an external 5-12V DC power supply. This stepper motor uses unipolar windings, as shown in Fig. 5. Terminal 5 Red is connected directly to 5-12V DC. Note how this connection goes to taps at the centre of both motor windings. At any one time, at most one of terminal 2 Pink or 4 Orange, but not both, should be driven to ground. Similarly, at any one time, at most one of 1 Blue or 3 Yellow, but not both, should be connected to ground. The three standard drive waveforms (wave drive, full step, and half step) that cause the motor's shaft to turn in steps are shown below in Fig. 6. The **28BYJ-48** stepper motor has 2048 full steps per revolution.



Figure 5: 28BYJ-48 unipolar stepper motor and coil windings schematic

Connect the DC power supply:

1. Make sure that the DC power supply is off.
2. Connect the ground terminal from the power supply to the ground rail (labeled "-") on the breadboard. Also connect the ground terminal on the driver module to ground on the breadboard.
3. Connect the Vcc terminal from the power supply to the power supply rail (labeled "+") on the breadboard. Also connect the positive supply terminal on the driver module to the power supply rail on the breadboard.
4. Carefully double-check all the wiring connections.
5. Turn on the power switch to the Zybo Z7 board. Then plug on the DC power supply.
6. Verify that the stepper motor can be operated using the provided stepper motor software.



ECE 315 Computer Interfacing

LAB 4: HARDWARE SETUP

Winter 2026



Figure 6: Timing diagram of step mode coil activation pattern